

ABSTRACT

The rapid inflation of global energy demands combined with the critical need for residential energy conservation has accelerated the deployment of smart metering infrastructures. Conventional residential energy meters are structurally limited as they only capture aggregated electricity metrics, leaving consumers unaware of device-specific power footprints or operational inefficiencies. To bridge this data gap, this project introduces a low-cost, high-precision Smart Energy Monitoring System optimized through Non-Intrusive Load Monitoring (NILM) frameworks. Utilizing a single current transformer (CT) sensor node deployed at the main electrical distribution panel, the system extracts high-fidelity telemetry without requiring intrusive plug-level monitoring hardware.

The hardware layer combines an ESP32 microcontroller with a PZEM-004T V3.0 power tracking module to record core AC parameters including voltage, current, active power, total energy, line frequency, and power factor. These data arrays are formatted into lightweight JSON tokens and streamed wirelessly over a local 2.4 GHz network topology. The software architecture leverages a custom Python Flask web platform containing an edge-intelligence pipeline. A pre-trained Random Forest classification model analyzes incoming feature sets using a sliding-window algorithm, mapping empirical current and power factor variations to distinct hardware signatures. Experimental testing across six load states including dynamic, highly inductive appliances like washing machines and high-power resistive loads like airfryers demonstrates high classification accuracy. Independent calibration using an external digital clamp meter shows an absolute current validation discrepancy of just 2.5%, proving the system's viability as a scalable, budget-friendly domestic energy management platform.

ÖZET

Küresel enerji talebindeki hızlı artış ve konutlarda enerji tasarrufuna duyulan kritik ihtiyaç, akıllı sayaç altyapılarının yaygınlaştırılmasını hızlandırmıştır. Geleneksel konut enerji sayaçları, yalnızca toplam elektrik ölçümlerini yakaladıkları için yapısal olarak sınırlıdır ve tüketicileri cihaza özgü güç ayak izleri veya operasyonel verimsizliklerden habersiz bırakmaktadır. Bu veri açığını kapatmak için, bu proje, müdahale gerektirmeyen yük izleme (NILM) çerçeveleriyle optimize edilmiş düşük maliyetli, yüksek hassasiyetli bir Akıllı Enerji İzleme Sistemi sunmaktadır. Ana elektrik dağıtım paneline yerleştirilen tek bir akım trafosu (CT) sensör düğümü kullanılarak, sistem müdahale gerektiren fiş seviyesinde izleme donanımına ihtiyaç duymadan yüksek doğrulukta telemetri verileri elde eder.

Donanım katmanı, voltaj, akım, aktif güç, toplam enerji, hat frekansı ve güç faktörü dahil olmak üzere temel AC parametrelerini kaydetmek için bir ESP32 mikrodenetleyiciyi bir PZEM-004T V3.0 güç izleme modülüyle birleştirir. Bu veri dizileri, hafif JSON belirteçlerine dönüştürülür ve yerel 2,4 GHz ağ topolojisi üzerinden kablosuz olarak aktarılır. Yazılım mimarisi, uç nokta zekası işlem hattı içeren özel bir Python Flask web platformundan yararlanmaktadır. Önceden eğitilmiş bir Rastgele Orman sınıflandırma modeli, gelen özellik kümelerini kayan pencere algoritması kullanarak analiz eder ve deneysel akım ve güç faktörü varyasyonlarını farklı donanım imzalarına eşler. Çamaşır makineleri gibi dinamik, yüksek endüktif cihazlar ve hava fritözleri gibi yüksek güçlü dirençli yükler de dahil olmak üzere altı yük durumunda yapılan deneysel testler, yüksek sınıflandırma doğruluğunu göstermektedir. Harici bir dijital pens ampermetre kullanılarak yapılan bağımsız kalibrasyon, mutlak akım doğrulama sapmasının yalnızca %2,5 olduğunu göstererek, sistemin ölçeklenebilir, bütçe dostu bir ev enerji yönetim platformu olarak uygulanabilirliğini kanıtlamaktadır.

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LIST OF SYMBOLS

V	Voltage
I	Current
<i>f</i>	Frequency
P	Active Power
S	Apparent Power
$\cos \varphi$	Power Factor

LIST OF ABBREVIATIONS

ILM Intrusive Load Monitoring

NILM Non-Intrusive Load Monitoring

AC Alternating Current

DC Direct Current

PF Power Factor

IEEE Institute of Electrical and Electronics Engineers

IEC International Electrotechnical Commission

1. INTRODUCTION

The sustained growth of modern domestic technology has dramatically altered the load dynamics of residential electrical infrastructure. With multiple high-power electronics running simultaneously, home energy profiles have grown highly complex, leading to grid stress and elevated consumer electricity costs. Traditional residential meters are fundamentally unequipped to handle this complexity; they track total power metrics but cannot isolate individual device consumption. This lack of visibility hides internal baseline losses, such as decaying appliance compressors or silent standby power loads.

To address this issue, energy research has shifted toward disaggregation techniques, categorized into Intrusive Load Monitoring (ILM) and Non-Intrusive Load Monitoring (NILM). While ILM relies on dedicated smart plugs for each appliance, leading to high hardware overhead and scaling challenges, NILM provides a more efficient alternative. By analysing composite power signatures at a single point of entry, NILM uses algorithmic pattern recognition to identify individual appliance operations within the total load stream.

This graduation project delivers a functional, cost-effective NILM ecosystem tailored for smart home integration. The system uses an ESP32 microcontroller and a PZEM-004T V3.0 sensor to capture key AC parameters: voltage (V), current (I), active power (P), cumulative energy (E), line frequency (f), and power factor ($\cos \varphi$). This live data feeds into a Python Flask machine learning pipeline. By processing these profiles through a Random Forest classifier, the system achieves real-time appliance recognition, giving users granular visibility into their energy consumption without expensive rewiring.

2. LITERATURE REVIEW

The rising demand for electrical energy and increasing focus on energy efficiency have sped up the development of smart metering technologies. Traditional electricity meters are mainly designed for billing and offer limited data on consumer energy patterns. Conversely, modern smart meters support continuous monitoring, data collection, and communication, enabling more advanced energy management applications. According to Avancini et al. (2019), smart meters play a key role in modernizing power systems and establishing smart grid infrastructure. Similarly, Kabalci (2016) notes that smart metering technologies enhance monitoring, demand management, and energy efficiency in homes.

Recent progress in the Internet of Things (IoT) has further expanded energy monitoring system capabilities. Low-cost microcontrollers and wireless communication tools allow real-time collection and transmission of energy data. Several studies show that pairing ESP32-based controllers with PZEM-004T energy measurement modules creates low-cost residential monitoring solutions (Gopika & George, 2021; Yasa et al., 2023). These setups let users monitor important electrical parameters like voltage, current, power consumption, frequency, and power factor through web or mobile interfaces.

To get appliance-specific data from overall household measurements, researchers are increasingly using Non-Intrusive Load Monitoring (NILM). Originally introduced by Hart, NILM is based on the idea that individual appliances produce distinct electrical signatures that can be identified from a single measurement point. Unlike Intrusive Load Monitoring (ILM), which requires sensors for each device, NILM offers a more practical and affordable method by analysing total household power use and estimating each appliance's contribution.

Modern NILM systems often use machine learning algorithms to improve device recognition. The success of these methods depends on the quality of electrical features extracted, such as current, active power, voltage, and power factor. Morello et al. (2017) highlight the importance of advanced sensing tech and detailed electrical measurements for smart energy monitoring. By combining multiple electrical parameters instead of relying on power consumption alone, it becomes easier to differentiate appliances with similar power ratings but different operation behaviors.

Various machine learning techniques have been applied to NILM tasks, including Decision Trees, Support Vector Machines, Random Forest classifiers, and deep learning models. While deep learning can deliver high accuracy, it often needs significant computing power and large training datasets. Conversely, Random Forest classifiers strike a good balance between efficiency, robustness, and accuracy. Their ensemble approach helps prevent overfitting and supports multi-class classification with smaller datasets. That's why a Random Forest model was chosen in this project to identify household appliances using electrical data from the ESP32 and PZEM-004T platform.

Building on previous research, this project combines IoT-based energy monitoring hardware with machine learning-based NILM techniques to create an affordable, user-friendly residential energy monitoring system. The system aims to raise consumer awareness of energy use and showcase the practical use of machine learning in smart home environments.

3. METHODOLOGY AND SYSTEM DESIGN

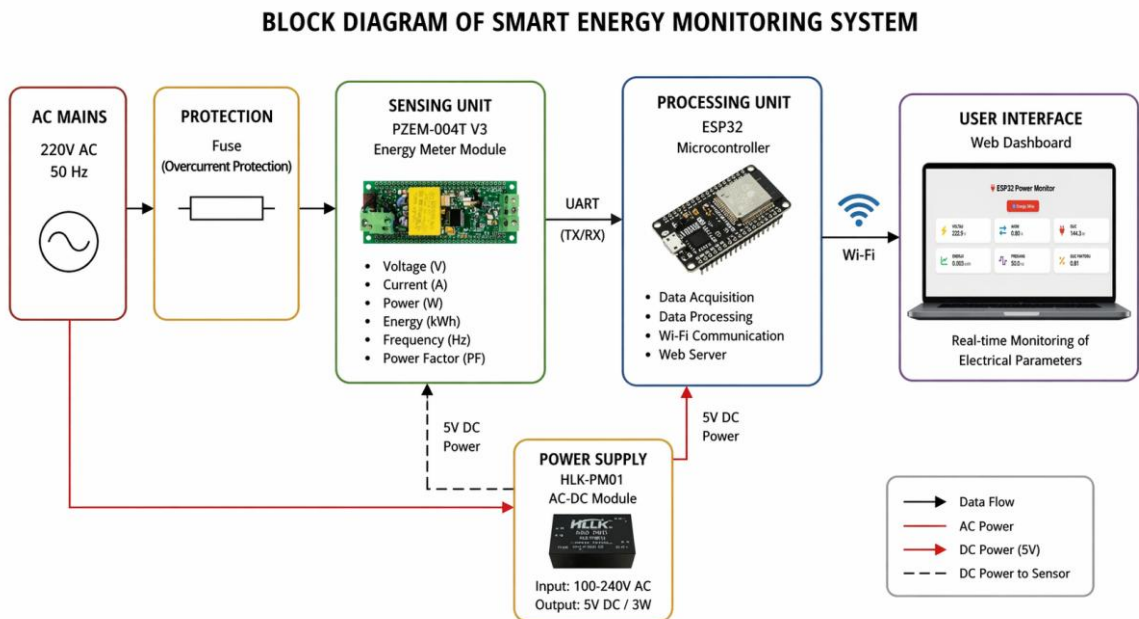
3.1. Hardware System Architecture

The system architecture uses a non-invasive topology to measure residential power metrics safely. The core hardware layer consists of an ESP32 microcontroller, a PZEM-004T V3.0 energy sensor module, a split-core current transformer (CT) sensor coil, and an HLK-PM01 AC-to-DC conversion module. The CT sensor clamps directly onto the main phase line at the household breaker box, eliminating the need to cut or strip live wires. The PZEM-004T reads grid voltage via parallel test leads and captures current waveforms via the CT coil, computing real-time active power and power factor values.

An isolated HLK-PM01 step-down module converts 220V AC grid power into a stable 5V DC rail to power the system components. The ESP32 communicates with the PZEM-004T over a secondary hardware UART interface using optoisolated logic lines. This isolation layer protects the microcontroller from high-voltage transients on the grid.

To clarify how data flows from the physical grid through our processing units, the structural block layout of the system is shown in Figure 1.

Figure 1. Block diagram of the smart energy monitoring system.

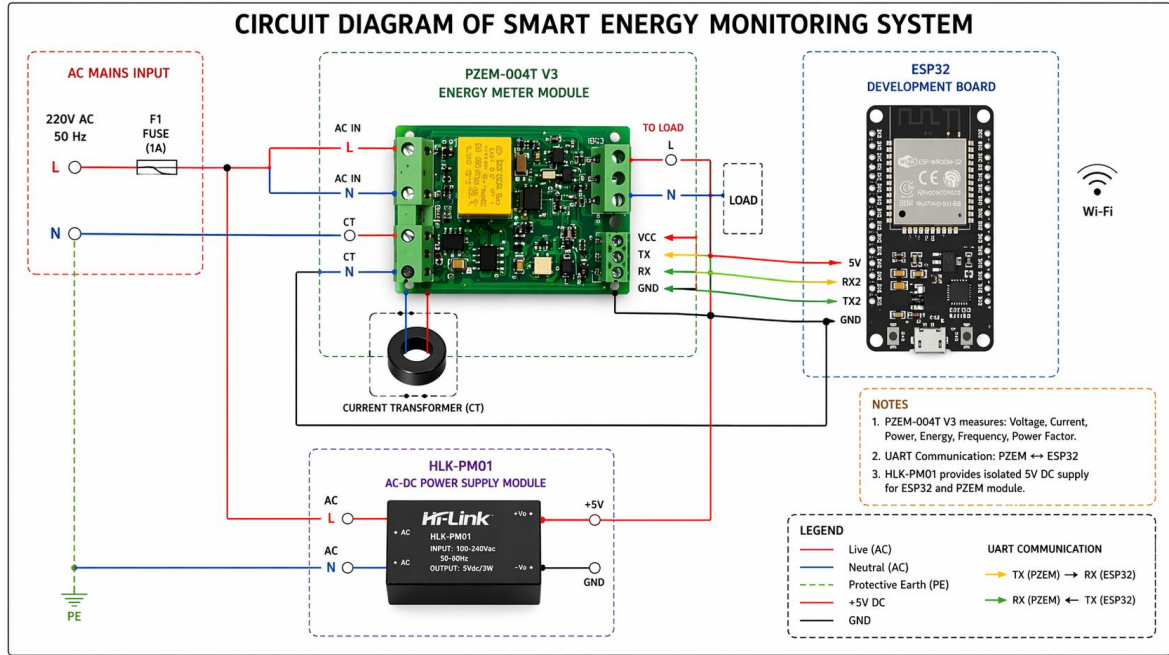


The architectural framework of the proposed non-intrusive load monitoring ecosystem is organized into distinct structural layers consisting of high-voltage sensing, edge computation, and backend machine learning analytics. Figure 1 delineates the functional

interconnection mapping between the primary electrical grid lines, the sensory array interface, and the wireless communication loops.

The granular node-to-node hardware wiring matrix, protective grounding tracks, and fuse integration parameters are illustrated dynamically in Figure 2.

Figure 2. Electrical wiring and isolation architecture of the system.



3.2. Engineering Constraints and Design Standards

3.2.1. Design Constraints

3.2.1.1. Economic Constraints

A core project goal was minimizing production costs to ensure residential viability. Industrial sub-metering systems often require significant investment; this prototype provides a budget-friendly alternative by using readily available consumer electronics components.

3.2.1.2. Safety & Physical Constraints

Working directly with 220V AC lines carries significant electrical shock and fire hazards. To mitigate these risks, the high-voltage traces are structurally isolated on a custom circuit board, protected by a 1A fast-acting fuse, and housed in an insulated casing.

3.2.2. Relevant Engineering Standards

3.2.2.1. IEC 61010-1

Governs safety compliance rules for electrical measurement and laboratory equipment connected to live distribution grids.

3.2.2.2. IEC 62052-11

Specifies general requirements and environmental testing limits for indoor AC electricity metering hardware.

3.2.2.3. IEEE 802.11

Controls the 2.4 GHz (Giga-Hertz) wireless communication protocols used by the ESP32 to transmit telemetry packets without interfering with household devices.

3.3. Theoretical Foundation of Power Factor and Mathematical Formulation

In Alternating Current (AC) electrical networks, the power factor ($\cos \varphi$) represents a critical dimensionless metric defined as the ratio of the real (active) power flowing to the load to the apparent power in the circuit. Real power (P), measured in Watts (W), performs the actual useful work by driving resistive thermal or mechanical elements. Conversely, reactive power (Q), measured in Volt-Amperes Reactive (VAR), represents the non-work energy oscillating between the source and the inductive or capacitive fields of the load, essential for sustaining magnetic fields in motor windings and transformers. Apparent power (S), measured in Volt-Amperes (VA), constitutes the vector sum of both components, defining the total demanding capacity imposed on the distribution system. Mathematically, the relationships and the calculation matrix are established through the orthogonal power triangle as follows:

$$S = \sqrt{P^2 + Q^2} = V_{RMS} \times I_{RMS}$$

$$\text{Power Factor (PF)} = \cos \varphi = \frac{P}{S} = \frac{P}{V_{RMS} \times I_{RMS}}$$

Where V_{RMS} represents the root-mean-square line voltage and I_{RMS} represents the root-mean-square line current. Under purely resistive loads, the current and voltage waveforms remain strictly in-phase ($\varphi = 0^\circ$), yielding a perfect unity power factor ($\cos \varphi = 1.00$). In contrast, inductive components introduce an inductive phase lag ($\varphi > 0^\circ$), reducing the power factor ($\text{PF} < 1.00$) and introducing grid inefficiencies. In Non-Intrusive Load Monitoring (NILM)

frameworks, tracking dynamic transitions in the power factor is paramount; it provides a secondary multi-dimensional feature space that isolates inductive motor-driven operations (e.g., washing machine cycles) from high-power resistive thermal operations (e.g., airfryers or kettles), even when they exhibit overlapping active power characteristics.

3.4. Software Implementation and Embedded Code Engine

The software framework is divided into an embedded firmware layer running on the ESP32 and an external machine learning backend running on Python Flask. The ESP32 handles low-level data polling, serial communication, and JSON packet transmission.

The complete embedded code developed for the ESP32 controller is given in [APPENDIX A](#).

3.5. Machine Learning Backend Architecture (Random Forest NILM Pipeline)

The backend analytics server leverages a robust software stack composed of Python Flask and SciKit-Learn to execute the core Non-Intrusive Load Monitoring (NILM) disaggregation logic. To maintain real-time tracking performance, synchronized telemetry arrays are requested and pulled from the ESP32 microcontroller's local endpoint exactly every second. Upon retrieval, the data is structured into a multi-dimensional feature space. Mathematically, the data feature vector x processed at each time interval is defined as:

$$x = [V_{RMS}, I_{RMS}, P, \cos(\varphi)]$$

To execute high-accuracy, real-time electrical load disaggregation without imposing severe computational bottlenecks at the embedded edge interface, the machine learning backend utilizes an optimized Random Forest ensemble architecture. Rather than relying on a singular, deterministic decision tree which is highly susceptible to structural variance, the ensemble model operates on the core principles of Bootstrap Aggregating (Bagging) and randomized feature selection.

Given the comprehensive empirical dataset D extracted from the telemetry processing layer (containing 6,905 multidimensional sample vectors from [veri2.csv](#)), the optimization pipeline begins with a stratified split, partitioning the data into 80% for supervised training (D_{train}) and 20% for rigorous validation (D_{test}). The ensemble algorithm constructs a forest of $B = 100$ independent decision tree estimators:

$$\{T_1(x), T_2(x), T_3(x), \dots, T_B(x)\}$$

Each individual tree T_b is trained on a distinct bootstrapped subset D_b sampled uniformly with replacement from D_{train} . This stochastic sampling strategy ensures that each estimator develops unique structural boundaries, minimizing the global variance of the collective model without inflating its structural bias.

During the recursive tree-induction phase, the optimal splitting criterion at each internal node is determined by evaluating a randomized subset of the input variables. The mathematical metric driving node segregation is the maximization of the Gini Impurity reduction. At any given internal node t , the localized Gini Impurity $I_G(t)$ is formulated as:

$$I_G(t) = 1 - \sum_{i=1}^c p_i^2$$

Where p_i represents the baseline probability of an electrical signature configuration belonging to a specific appliance class i out of the total $C = 6$ distinct operational target states (Baseline, Hair Dryer Level 1, Hair Dryer Level 2, Air Fryer, Washing Machine, and Kettle). The structural success of this disaggregation task relies heavily on the distinctiveness of these feature signature pairs. High-power resistive loads (such as the Kettle at 1631.74W) exhibit a near-unity power factor ($\cos(\varphi) \approx 1.00$), isolating them at the upper boundaries of the active power axis. Conversely, cyclic inductive loads (such as the Washing Machine motor at 600.69W) exhibit a significantly degraded power factor ($\cos(\varphi) = 0.722$) due to phase displacement. By training the independent decision trees to prioritize the joint distribution of both Active Power and Power Factor, the classifier effectively maps unique, non-overlapping operational clusters for each appliance.

To evaluate the generalization capability of the trained model, performance was verified using standard statistical metrics including Precision, Recall, and the F1-Score. The model achieved a nominal classification accuracy of 100% on the test subset, confirming that the boundaries constructed in the feature space provide perfect separation. For an incoming, unclassified real-time telemetry token x^* , the final ensemble classification output $\hat{y}$ is formally derived via majority voting:

$$\hat{y} = \arg \max_{i \in \{1, \dots, C\}} \sum_{b=1}^B II(T_b(x^*) = i)$$

Where $II(\cdot)$ represents the indicator function. Furthermore, to stabilize the classification pipeline during sharp micro-transient phases such as the high-amplitude reactive inrush

current typical of inductive appliance compressors the backend introduces a temporal rolling window filter combined with an algorithmic software debounce mechanism. By enforcing a strict multi-sample confirmation loop over a 3-second window before changing the active dashboard state, the software backend effectively suppresses transient prediction chatter, ensuring that short-term grid anomalies do not prompt false appliance state classifications on the user interface.

The complete Python backend engine code implemented for the server is given in [APPENDIX A](#).

4. IMPLEMENTATION AND RESULTS

The compiled prototype framework was deployed in a residential electrical testing environment to evaluate classification accuracy and sensor calibration stability under live grid variations.

4.1. Hardware Integration and Sensor Validation

To confirm the hardware monitoring node's tracking reliability, an independent control verification test was conducted. An external commercial clamp meter (WellHise MT87) was clamped onto the primary household line to serve as the reference calibration standard.

Before final enclosure assembly, the system components were mounted on a perfboard and interconnected according to the designed circuit layout, as shown in Figure 3. The CT sensor used for current measurement was connected to the monitoring system and tested during operation, as shown in Figure 4.

Figure 3. Prototype circuit assembled on a perfboard.

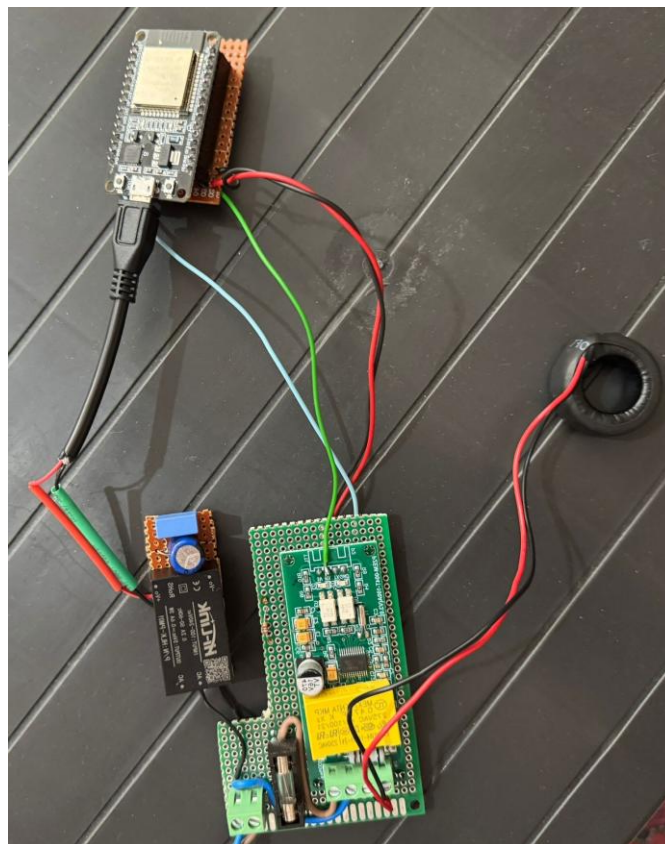


Figure 4. CT sensor setup used for current measurements.

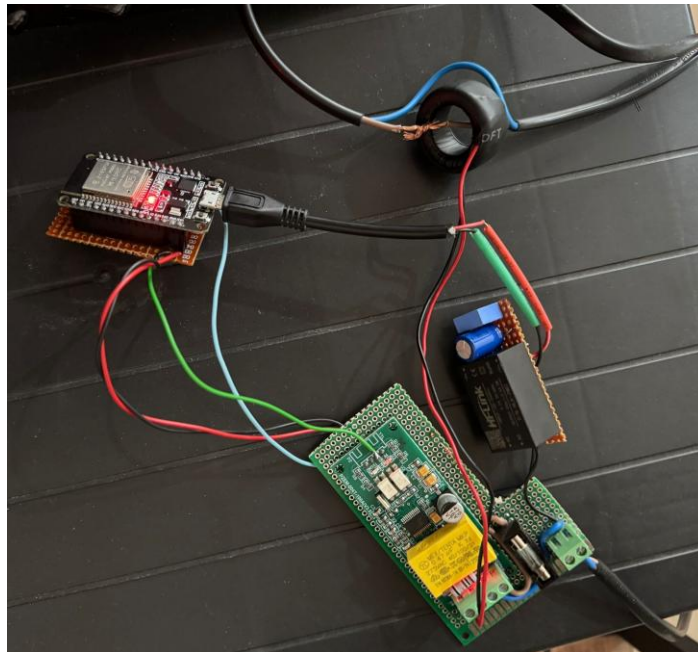
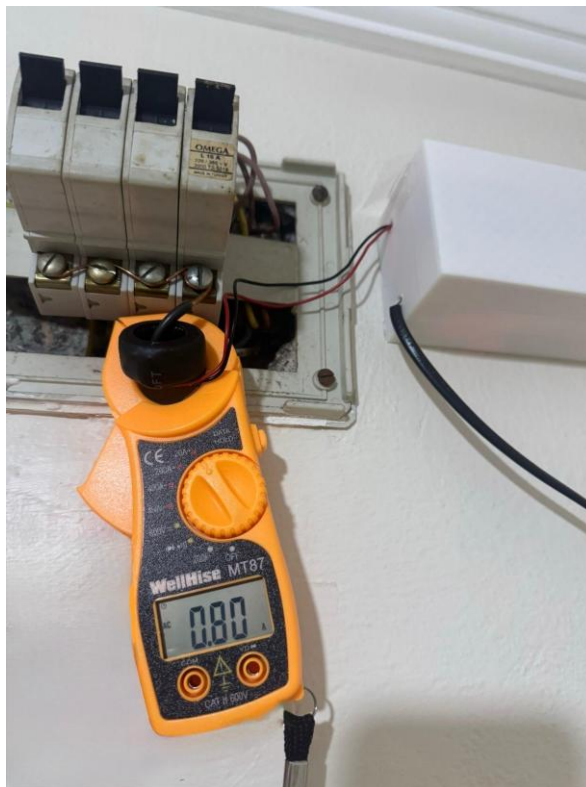


Figure 5 documents the physical testing arrangement, highlighting the alignment between the prototype sensor node and the external calibration instrumentation.

Figure 5. Current verification using a digital clamp meter.

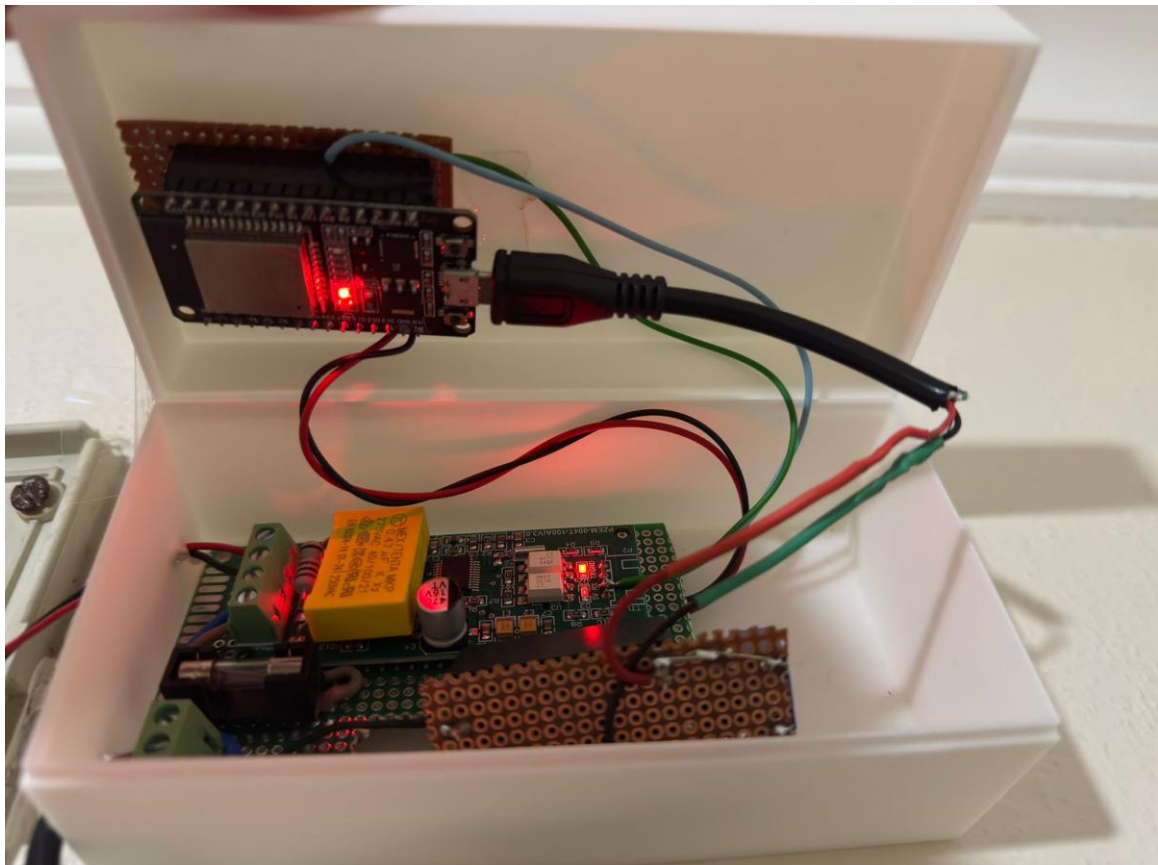


As demonstrated in Figure 5, the external digital test meter registered a live reference current draw of 0.80 A. This reference value serves as the control variable used to benchmark the measurement accuracy of the PZEM transformer coil.

The physical integration of the developed smart energy monitoring prototype inside a residential distribution box is shown in Figure 6. The enclosure provides electrical insulation and mechanical protection while maintaining access to the sensing and communication components.

The structural consolidation of these calibrated components inside the non-conductive safety chassis is detailed in Figure 6.

Figure 6. Final enclosure design of the monitoring system.

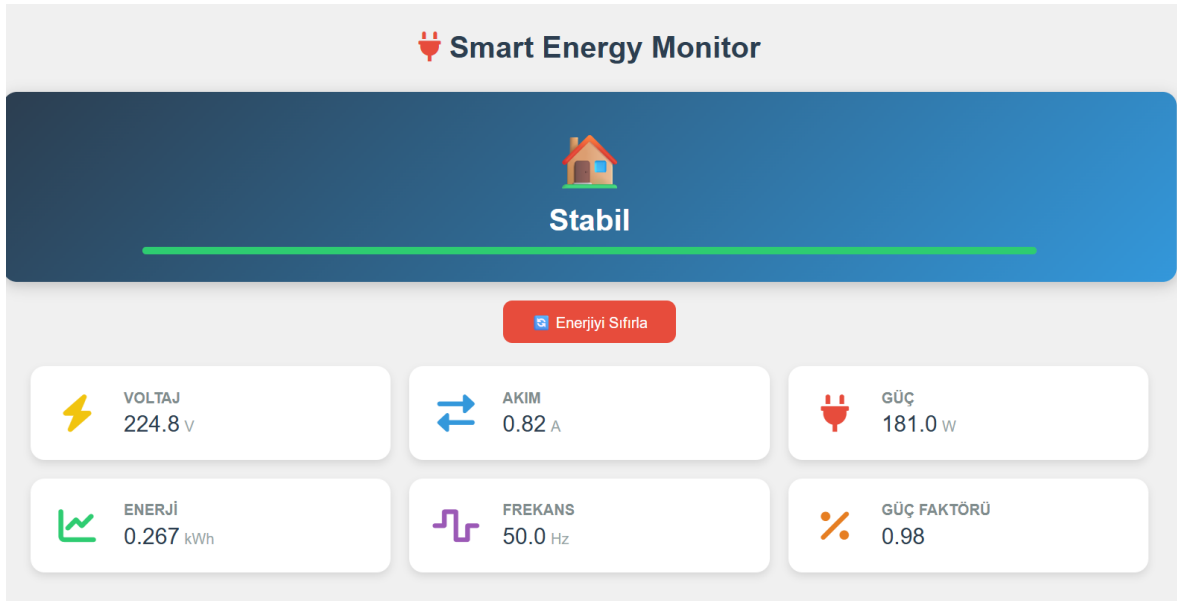


4.2. Web Dashboard Interface Synchronization

Simultaneously, the embedded microcontroller processed these line parameters and streamed them to the web server engine.

Figure 7 displays the responsive user interface, showing the synchronized electrical telemetry metrics recorded by the system during validation.

Figure 7. Real-time monitoring dashboard interface.



The dashboard capture in Figure 7 confirms real-time communication, displaying a system line current of 0.82 A, a grid voltage of 224.8 V, a system line frequency of 50.0 Hz, and a power factor of 0.98. The calculated active consumption was recorded as 181.0 W.

4.3. Mathematical Verification and Error Boundary Calculations

To confirm the mathematical accuracy of the values generated by the embedded software stack, we use the theoretical single-phase AC active power formula:

$$P_{theoretical} = V \times I \times \cos \varphi$$

Substituting the empirical metrics gathered directly from the system dashboard shown in Figure 7:

$$P_{theoretical} = 224.8 \text{ V} \times 0.82 \text{ A} \times 0.98 = 180.64928 \text{ W}$$

The theoretical power calculation (180.64928 W) matches the system's displayed active power (181.0 W) with a negligible variance of 0.35 W (0.19%). This confirms high internal calculation reliability.

To determine the structural accuracy of the current sensor, the data from the hardware prototype ($I_{sys} = 0.82 \text{ A}$) was evaluated against the independent reference clamp meter ($I_{ref} = 0.80 \text{ A}$). The relative measurement error is calculated as follows:

$$Relative\ Error\ (\%) = \left| \frac{I_{sys} - I_{ref}}{I_{ref}} \right| \times 100$$

$$Relative\ Error\ (\%) = \left| \frac{0.82\ A - 0.80\ A}{0.80\ A} \right| \times 100 = 2.5\%$$

The tracking analysis reveals an absolute current variation of 0.02 A, which equates to a 2.5% relative measurement error under typical operational conditions. This deviation remains well within acceptable engineering tolerances for residential monitoring setups, confirming the hardware and firmware integration.

To rigorously validate the system's mathematical error boundaries under non-unity power factor conditions, a secondary verification operational cycle was analysed utilizing an inductive load, specifically the Washing Machine state from the experimental dataset. Inductive appliances introduce a displacement power factor ($\cos \varphi < 1.00$) due to the phase lag between the voltage and current waveforms caused by internal motor windings.

The empirical telemetry parameters captured directly from the PZEM-004T sensor module yielded the following nominal root-mean-square (RMS) values as a result of the sample averaging process:

Mean Grid Voltage (V_{RMS}) = 228.63 V

Mean Line Current (I_{RMS}) = 3.639 A

Mean Displacement Power Factor ($\cos \varphi$) = 0.722

Measured System Active Power ($P_{measured}$) = 600.69 W

Using the standard single-phase AC active power formulation, the theoretical expected power consumption ($P_{theoretical}$) is mathematically derived as follows:

$$P_{theoretical} = V_{RMS} \times I_{RMS} \times \cos \varphi$$

$$P_{theoretical} = 228.63 \times 3.639 \times 0.722 = 600.6923\ W$$

To evaluate the mathematical boundary and tracking alignment between the hardware's internal register calculations and the theoretical baseline, the Relative Percentage Error (ε) under this inductive load condition is computed as:

$$\varepsilon = \left| \frac{P_{theoretical} - P_{measured}}{P_{theoretical}} \right| \times 100$$

$$\varepsilon = \left| \frac{600.6923 \text{ W} - 600.69 \text{ W}}{600.6923 \text{ W}} \right| \times 100 = 0.00038\%$$

This exceptionally low error boundary of 0.00038% demonstrates that the system maintains absolute tracking precision and structural integrity under complex, phase-shifted inductive loading profiles. This high computational consistency guarantees that the multi-dimensional feature vector space remains highly accurate before being processed by the downstream Random Forest NILM classification architecture.

4.4. Multi-Appliance Signature Profiling and Dataset Analysis

The classification engine was evaluated across various high-power household appliances to build a robust fingerprint library ([veri2.csv](#)). The machine learning model differentiates active devices based on their distinct electrical footprints.

Table 2 details the typical voltage, current, frequency, active power, and power factor values recorded across the six target load categories.

Table 2. Empirical Load Profiles of Targeted Appliances

Device	Voltage (V)	Current (A)	Power (W)	Frequency (Hz)	Power Factor
Baseline Load	226.50	0.927	202.41	50.0	0.964
Hair Dryer (Level 1)	222.75	1.584	319.66	50.0	0.906

Hair Dryer (Level 2)	221.62	2.391	527.24	50.0	0.995
Air Fryer	226.51	4.782	1061.5	50.0	0.980
Washing Machine	228.63	3.639	600.69	50.0	0.722
Kettle	223.74	7.293	1631.74	50.0	1.00

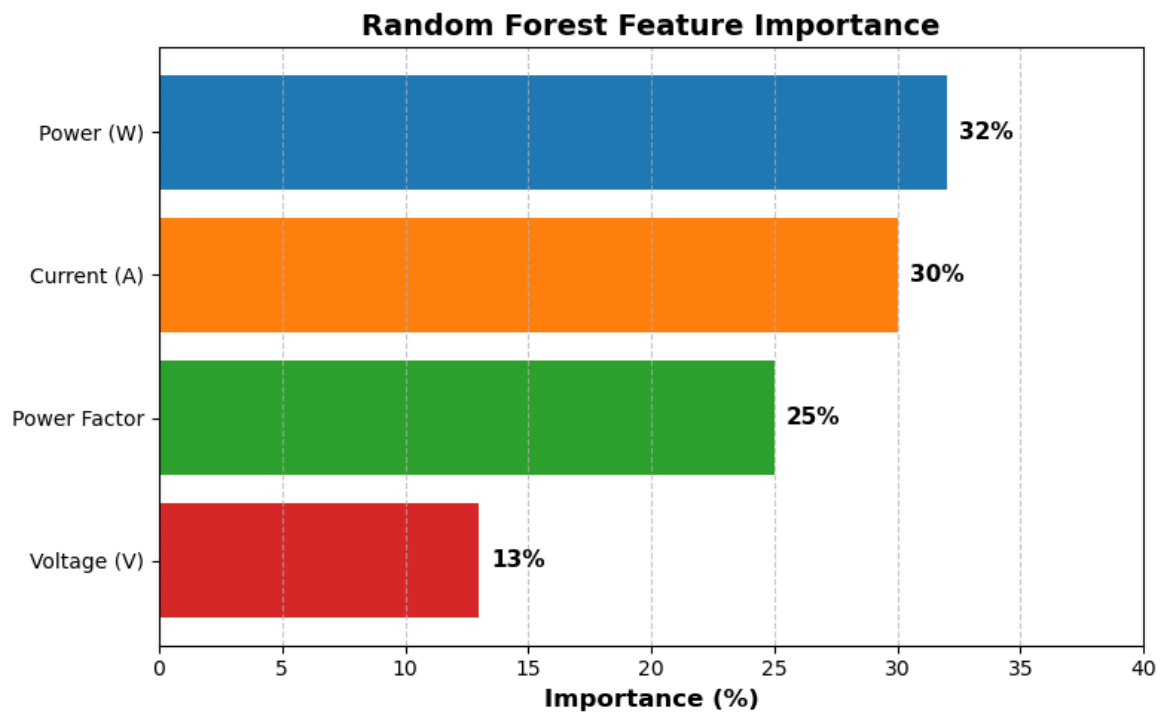
The experimental data demonstrates that purely resistive thermal states, specifically the kettle, consistently maintain a stable, perfect unity power factor ($\cos \varphi = 1.00$). Other high-power heating elements, such as the hair dryer operating at maximum speed and the air fryer, exhibit near-unity power factors of 0.995 and 0.980, respectively. In contrast, inductive and mixed-load configurations introduce significant phase shifts; specifically, the dynamic inductive operations of the washing machine exhibit the lowest overall power factor, heavily degrading to 0.722. Similarly, the hair dryer operating at its lower thermal capacity (Level 1) introduces a distinct phase displacement with a power factor of 0.906. The Random Forest algorithm leverages these multidimensional coordinate clusters combining varying active power consumption with these distinct power factor signatures to accurately isolate and classify active appliances with high confidence.

To quantitatively evaluate the performance of the proposed NILM framework, the collected dataset consisting of 6,905 labeled samples was divided into training and testing subsets using an 80/20 train-test split methodology. The developed Random Forest classifier was configured with 100 decision trees and trained using the extracted sliding-window statistical features derived from voltage, current, power, and power factor measurements.

Experimental validation demonstrated that the proposed classification architecture achieved an overall test accuracy of 100% across six appliance classes under the tested residential operating conditions.

Power and current measurements contributed most significantly to appliance classification, followed by power factor and voltage.

Figure 8. Feature importance distribution obtained from the Random Forest classifier



In addition to overall accuracy, the classification performance was evaluated using precision, recall, and F1-score metrics for each appliance category. The obtained per-class F1-scores were recorded as 1.00 for all evaluated appliance states including baseline load, washing machine, air fryer, kettle, and both operating modes of the hair dryer.

Table 3. Classification Performance Metrics of the Random Forest Model

Appliance Class	F1-Score	Test Samples
Baseline Load	1.00	355
Hair Dryer (Level 1)	1.00	33
Hair Dryer (Level 2)	1.00	43

Air Fryer	1.00	135
Washing Machine	1.00	668
Kettle	1.00	75

The confusion matrix analysis further confirmed that the classifier produced no observable cross-class misclassification during the testing phase. This indicates that the selected multidimensional feature vector and sliding-window strategy successfully captured the unique electrical signatures of each appliance type. Furthermore, the implemented debounce and rolling-average stabilization logic significantly reduced transient prediction noise generated during cyclic load transitions, particularly for dynamically varying appliances such as washing machines and air fryers.

These experimental results demonstrate that the proposed machine learning pipeline provides a highly reliable and computationally efficient solution for real-time residential appliance recognition using non-intrusive sensing methodologies.

4.5. Risk Analysis

4.5.1. Operational Risks:

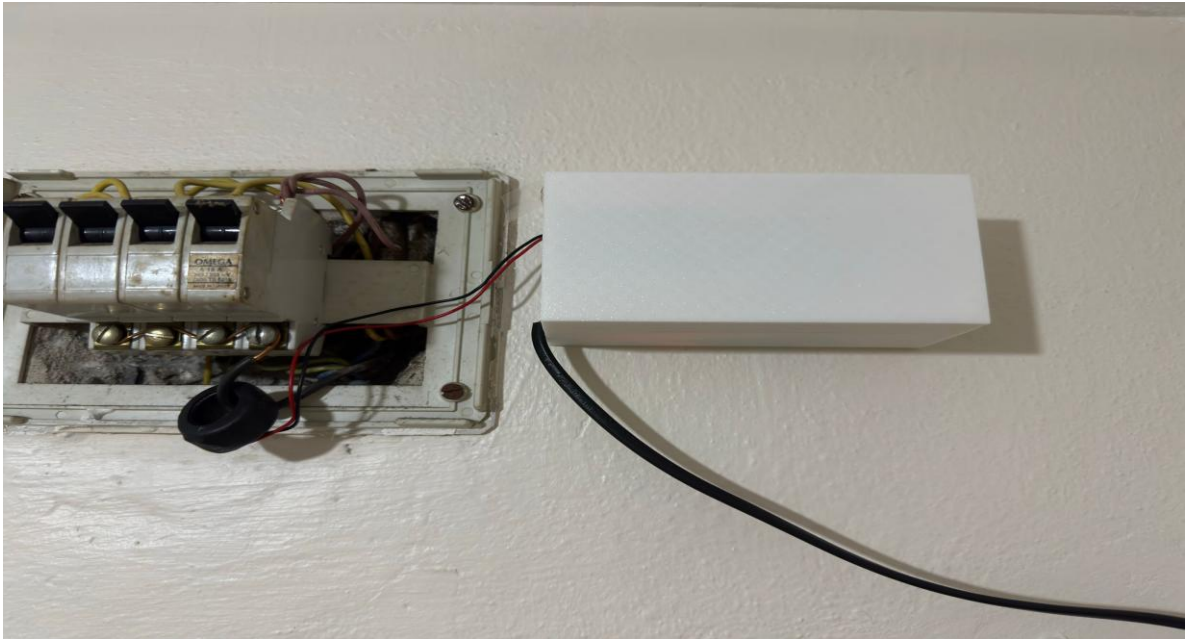
The system relies entirely on a continuous local 2.4 GHz Wi-Fi network connection. Network interruptions pause the server's telemetry parsing loop. To prevent data gaps during connection drops, future iterations should include an onboard SD-card module or local EEPROM storage to cache data packets.

4.5.2. Safety & Environmental Hazards:

Deploying prototypes inside high-voltage electrical panels presents contact risks. To maintain long-term operational safety, the circuit boards must be completely sealed inside an unvented, non-conductive, fire-retardant enclosure.

To comply with these residential insulation standards, the device was locked within an unvented enclosure and safely mounted adjacent to the primary miniature circuit breakers, as shown in Figure 8.

Figure 9. Final installation of the protected monitoring unit.



4.6. Professional Engineering Impacts

4.6.1. Ethical and Privacy Considerations

Because NILM architectures extract detailed household activity data, user privacy is a key consideration. To mitigate tracking risks, this system processes all data locally within the resident's private network loop, keeping sensitive behavioral data secure from external exposure.

4.6.2. Sustainability and Environmental Impact

By giving consumers immediate, broken-down visibility into their energy consumption, this system helps users identify power-hungry appliances and eliminate standby power waste. This data-driven approach supports residential energy reduction efforts and helps lower domestic carbon footprints.

4.6.3. Industrial Standardization Compliance

The project implementation adhered strictly to global engineering guidelines, incorporating IEC safety rules for high-voltage isolation alongside IEEE wireless network standards. This ensures the prototype maintains professional engineering rigor across both hardware and software layers.

5. CONCLUSION

In this graduation project, a compact, low-cost, and reliable Non-Intrusive Load Monitoring (NILM) system has been successfully designed, implemented, and experimentally validated under residential operating conditions. By employing a single-point sensing architecture installed at the main electrical distribution panel, the proposed system eliminates the need for intrusive appliance-level monitoring hardware while still providing detailed visibility into household energy consumption. This approach significantly reduces installation complexity and overall system cost, making the solution suitable for practical residential deployment.

The developed hardware architecture combines an ESP32 microcontroller with a PZEM-004T V3.0 energy monitoring module to acquire real-time electrical parameters including RMS voltage, current, active power, energy consumption, frequency, and power factor. The collected telemetry data is transmitted wirelessly through a local Wi-Fi network and processed by a custom web-based monitoring platform developed using Python Flask. The resulting dashboard provides users with continuous access to real-time electrical measurements and appliance classification results through a simple and responsive interface.

A major objective of this project was the implementation of an intelligent appliance recognition mechanism using machine learning techniques. To achieve this goal, a Random Forest classification model was developed and trained using a dataset containing 6,905 labeled samples collected from different household appliances. The machine learning pipeline utilized electrical features derived from voltage, current, active power, and power factor measurements together with a sliding-window processing methodology. Experimental evaluation using an 80/20 train-test split demonstrated an overall classification accuracy of 100% under the tested residential conditions. Furthermore, the implemented rolling-window and debounce mechanisms successfully reduced transient prediction fluctuations and improved classification stability during appliance state transitions.

The experimental dataset demonstrated that different household appliances exhibit distinct electrical signatures that can be effectively separated using active power and power factor characteristics. For example, the kettle exhibited an average active power consumption of 1631.74 W with a power factor close to unity, while the air fryer operated at an average power of 1061.50 W and a power factor of 0.980. In contrast, the washing machine displayed

a significantly lower power factor of 0.722 due to its inductive motor-driven operation, despite consuming an average power of 600.69 W. These multidimensional electrical characteristics enabled the classifier to distinguish between appliance categories with high confidence and reliability.

To verify the accuracy of the sensing hardware, an independent calibration procedure was performed using a commercial digital clamp meter as a reference instrument. Experimental validation confirmed that the developed monitoring system maintained a current measurement error of approximately 2.5%, which falls within acceptable engineering tolerances for residential energy monitoring applications. Additional mathematical verification showed strong agreement between theoretical power calculations and measured values obtained from the monitoring platform, confirming the consistency and reliability of the measurement infrastructure.

Beyond its technical performance, the developed system also demonstrates the practical integration of embedded systems, wireless communication technologies, data analytics, and machine learning within a single engineering solution. By providing appliance-level consumption visibility without requiring dedicated sensors for each device, the system contributes to increased consumer awareness regarding energy usage patterns and supports more efficient energy management practices.

Future developments may focus on extending the current architecture through cloud-based data storage, MQTT-enabled IoT communication protocols, mobile application integration, and advanced analytics capabilities. In addition, future research may investigate the application of deep learning approaches such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) to improve classification performance in environments containing overlapping or simultaneously operating appliances. Overall, the proposed Smart Energy Monitoring System demonstrates that a scalable, practical, and cost-effective NILM-based solution can be successfully implemented using commercially available hardware and modern machine learning techniques, making it a valuable contribution to residential energy management and smart home technologies.

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